

MISHIMA, Japan

WMO: 476570

Lat: 35.1139N

Lon: 138.9253E

Elev: 72

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	30.4	10.6	9.5	43.1	15.9	12.3	41.2	19.6	48.8	18.0	49.5	2.4	120	0.370

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.2	91.5	77.8	89.8	77.2	88.0	76.4	79.6	87.9	78.6	86.5	77.7	85.3	8.5	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	142.3	84.3	76.3	137.7	83.6	75.4	133.3	82.9	43.2	88.0	42.2	86.7	41.3	85.5	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.2	14.5	12.3	DB	24.0	95.1	1.9	1.4	22.6	96.2	21.4	97.0	20.3	97.8	18.9	98.9	(4)
(5)			WB	21.4	80.6	2.0	1.2	20.0	81.4	18.8	82.1	17.7	82.8	16.2	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.9	42.7	44.3	49.9	58.3	66.4	72.4	79.8	81.9	75.9	66.1	56.1	47.2	(6)	
	DBStd	14.19	4.08	4.95	5.49	5.57	4.05	4.01	4.15	3.43	4.87	5.06	5.19	5.05	(7)	
	HDD50	603	230	173	72	5	0	0	0	0	0	0	7	116	(8)	
	HDD65	2850	691	580	470	208	32	1	0	0	1	46	270	551	(9)	
	CDD50	4932	4	13	68	256	510	673	923	990	776	500	189	30	(10)	
	CDD65	1701	0	0	0	8	76	224	458	525	327	81	2	0	(11)	
	CDH74	14098	0	0	1	21	288	1078	4258	5612	2484	351	5	0	(12)	
	CDH80	4566	0	0	0	1	16	175	1423	2186	720	45	0	0	(13)	
(14) Wind	WSAvg	4.9	4.5	4.7	5.4	5.5	5.3	5.0	5.3	5.3	4.8	4.3	4.0	4.4	(14)	
(15) Precipitation	PrecAvg	74.5	3.0	3.6	5.8	6.1	6.7	8.5	8.5	7.2	10.2	7.7	4.7	2.6	(15)	
	PrecMax	106.2	6.6	7.6	11.1	12.2	12.9	13.7	18.6	20.2	20.6	29.7	11.0	6.5	(16)	
	PrecMin	49.6	0.0	0.7	2.0	2.6	2.8	4.1	0.9	0.7	2.9	0.8	0.4	0.1	(17)	
	PrecStd	13.9	1.8	1.9	2.2	2.5	2.4	2.6	4.6	4.9	5.1	5.6	2.6	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	65.9	70.4	76.7	82.3	87.3	93.2	94.4	91.1	84.4	74.0	67.2	(19)	
		MCWB	52.6	55.8	56.3	61.7	67.1	74.3	77.9	78.9	76.7	70.8	63.1	59.1	(20)	
	2%	DB	57.4	61.0	66.6	73.3	79.2	84.3	90.8	92.1	88.6	80.7	71.2	63.1	(21)	
		MCWB	48.4	51.9	55.7	60.5	65.1	72.2	77.6	78.3	75.4	68.9	60.8	54.0	(22)	
	5%	DB	54.7	57.4	63.6	71.1	77.2	81.8	89.0	90.6	86.6	77.7	68.7	60.0	(23)	
		MCWB	45.8	48.4	53.6	59.7	64.7	70.5	77.0	77.7	74.6	67.5	59.3	51.0	(24)	
	10%	DB	52.3	54.5	60.9	68.9	75.2	79.6	87.1	88.9	84.3	75.3	66.2	57.3	(25)	
		MCWB	43.5	45.5	51.6	58.4	63.8	69.8	76.4	77.1	73.6	65.6	57.8	49.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	58.4	61.1	68.1	71.9	76.7	80.4	81.1	78.9	74.3	66.5	61.8	(27)	
		MCDB	58.7	63.4	65.4	71.9	76.6	84.4	89.3	90.1	86.0	79.3	70.3	65.7	(28)	
	2%	WB	50.5	53.6	58.1	64.5	69.3	74.8	79.1	80.0	77.4	71.8	63.7	55.7	(29)	
		MCDB	55.4	59.4	63.6	69.3	74.7	81.0	87.4	88.6	84.4	78.1	68.3	60.3	(30)	
	5%	WB	46.9	49.8	56.0	62.1	67.5	73.3	78.2	79.1	76.4	69.5	61.4	52.7	(31)	
		MCDB	52.9	55.7	61.7	68.4	73.6	79.0	86.1	87.2	83.3	75.9	66.2	58.0	(32)	
	10%	WB	44.5	46.8	53.2	60.2	65.9	71.7	77.3	78.3	75.4	67.3	59.1	50.1	(33)	
		MCDB	50.9	52.9	59.2	66.6	72.4	77.7	84.8	86.0	82.1	73.2	64.5	56.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	17.7	17.6	17.1	15.7	12.8	12.1	13.2	14.1	15.5	17.0	17.9	(35)	
		MCDBR	20.4	20.7	20.5	19.5	18.2	15.4	14.5	15.0	15.9	17.5	19.2	20.0	(36)	
	5% WB	MCWBR	13.0	13.4	12.5	10.3	8.6	6.4	5.4	5.6	6.4	8.5	11.2	12.6	(37)	
		MCWBR	18.3	17.7	17.4	15.4	14.5	12.0	13.0	13.5	13.8	15.0	16.0	17.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.387	0.441	0.506	0.562	0.584	0.611	0.598	0.590	0.529	0.500	0.431	0.383	(40)	
		taud	2.213	2.064	1.911	1.828	1.835	1.837	1.918	1.926	2.052	2.051	2.159	2.244	(41)	
	Ebn at Noon	Ebn at Noon	251	248	242	236	233	226	228	226	234	227	232	243	(42)	
		Edh at Noon	35	46	58	66	66	66	61	59	50	46	37	32	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	881	1059	1320	1523	1640	1434	1580	1600	1262	999	827	778	(44)	
	RadStd	RadStd	58	96	85	159	170	134	237	134	111	105	73	59	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	+2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page